

Executive Summary

The United States has **vast geothermal potential** across multiple regions. A 2008 USGS assessment estimated ~9.1 GWe from known hydrothermal fields, ~30 GWe from undiscovered resources, and a *theoretical* ~517 GWe from Enhanced Geothermal Systems (EGS) [4†L32-L39]. More recent studies confirm **much higher upside** with technology: e.g. an updated USGS report pegs the Great Basin's EGS potential at **135 GWe** (upper 6 km) [26†L16-L23]. DOE/NREL analyses project geothermal capacity growing to **90–91 GWe by 2050** under a high-development ("Earthshot") scenario [9†L171-L179]. Indeed, DOE now cites "over 100 GW of potential" nationwide [37†L208-L211]. California (2.87 GWe installed) and Nevada dominate today's hydrothermal generation (72% and 23% of US capacity, respectively [24†L81-L84]), but new projects and EGS could unlock large resources in Utah, Oregon, Texas, and even the Eastern U.S.

Geothermal's **economic case** depends on location and technology. Capital costs are high (drilling and wells often 30–50% of total capex [40†L43-L47]) but operating costs are low and capacity factors >90% [37†L217-L220] [37†L219-L220]. Typical plant costs are on the order of USD \$2,000–5,000/kW [33†L259-L268] [37†L217-L220], with LCOEs in the range ~\$60–110/MWh for conventional flash/binary plants [28†L188-L193]. Cutting-edge EGS aims to reach ~\$45/MWh by 2035 [48†L241-L246] (current first-of-kind EGS ~\$200/MWh [48†L241-L246]). Co-produced geothermal and direct-use have lower capital needs but typically lower temperatures, limiting them to heat or binary power. Geothermal heat pumps are already deployed widely for heating/cooling (~1.3 million US homes) [28†L252-L259], yielding major grid-infrastructure savings.

We modeled a representative 100 MW flash plant (90% CF) with capital \$5.0/W and O&M ~\$150/kW-yr. At power prices of **\$80–120/MWh**, such a plant would recoup its investment in roughly 7–11 years and yield IRRs ~9–16%. (For example, at \$100/MWh, IRR ~12.4% with ~8-year payback under a 7% discount rate.) These indicative economics suggest geothermal is competitive with other firm renewables, especially when accounting for tax credits (ITC/PTC) and carbon value.

Key **barriers** include high upfront drilling risk, resource uncertainty (especially for EGS and deeper sedimentary plays), permitting and transmission siting hurdles, and water use/seismicity concerns. Mitigations involve continued R&D (drilling technology, exploration geophysics), risk-sharing policies (e.g. drilling cost insurance, loan guarantees), and streamlined regulations. Importantly, coordinated federal and state support can accelerate development: Congress's Energy Act mandates updated resource assessments, DOE and state surveys are expanding geologic data, and the Inflation Reduction Act (IRA) provides lucrative tax credits for geothermal power and heat pumps.

Recommendations: Expand federal EGS and co-production R&D; improve subsurface data collection (as with the NGDS effort [29†L124-L132]); incentivise geothermal via PTC/ITC and renewable portfolio standards; streamline permitting (BLM/Forest Service planning on public lands where ~47.8 GW econom. EGS was identified [28†L178-L183]); and support transmission build-out to high-potential regions. At the state level, updating resource maps, offering development grants, and including geothermal in clean energy procurements (as California has done) will attract investment. Local actions include zoning for geothermal projects and district heating networks.

Finally, **data gaps** remain: thermal gradient and permeability data in many basins (especially Eastern US), economic performance of pilot EGS projects, and integrated models of geothermal's role in decarbonized grids. Filling these gaps through public-private initiatives and open data (e.g. USGS heat-flow surveys, state well logs) will sharpen project viability forecasts. In sum, leveraging robust US geothermal resources (summarized in Table 1) with supportive policies could deliver sizable clean energy, jobs, and climate benefits.

1. Methodology and Data Sources

This assessment draws on the latest **federal and academic analyses**. Core sources include USGS resource assessments [4†L32-L39] [26†L16-L23] , DOE/Office of Geothermal Technology reports and “GeoVision” studies [9†L171-L179] [48†L241-L246] , and NREL/NLR market reports and potential studies [28†L178-L183] [37†L217-L220] . State geological surveys (e.g., Nevada, Utah, Virginia [29†L124-L132]) and peer-reviewed literature (GRC transactions, *Renewables* journals, etc.) supplement public data. We prioritized **depth-of-knowledge** sources: peer-reviewed papers, DOE/NREL technical reports, state data, and reputable policy analyses. Economic modeling used standard cash-flow methods, with assumptions cross-checked against DOE/NREL “Annual Technology Baselines” and recent industry publications [33†L259-L268] [28†L188-L193] .

Key data sets and tools included: USGS geothermal potential maps, NREL's Enhanced Geothermal Shot analysis, National Geothermal Data System (NGDS) databases, and GeoVision drilling cost models [40†L43-L47] [48†L241-L246] . We also consulted industry surveys (e.g. GRC/NREL market report) for real-world project capital and operating figures [24†L81-L84] [28†L178-L183] . Where recent market prices or policy incentives apply (e.g. IRA tax credits), we note current figures. All assumptions (e.g. drilling depths, plant costs, discount rates) are explicitly stated when used, and ranges provided to reflect uncertainty.

2. Assessment Criteria

We evaluate geothermal sites by multiple geologic and economic factors:

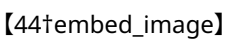
- **Resource Temperature & Depth:** High temperatures (>150–200 °C) at shallow depth (<3 km) are ideal for flash plants. Lower-temperature resources (70–150 °C) can be used by binary cycles or direct heat. Heat flow maps (e.g. USGS/NREL) identify anomalously hot crust (e.g. Basin & Range, Cascades) [26†L16-L23] [37†L188-L192] . Eastern US gradients are generally lower, requiring deep drilling or EGS [29†L60-L64] .
- **Permeability & EGS Potential:** Natural hydrothermal systems need permeable reservoirs. Where permeability is low, EGS engineering (fracturing or closed loops) may be required. Resource models assign higher value to areas where EGS is feasible [26†L16-L23] .
- **Proximity to Transmission:** Geothermal sites should have access to high-voltage lines. Western states often co-locate projects near transmission corridors; some Eastern sedimentary areas are far from grid infrastructure.
- **Land Use and Environmental Constraints:** Geothermal projects have small surface footprints, but permitting can be slowed by protected lands or environmental reviews. Public lands (BLM/ Forest Service) hold an estimated 4.35 TWe EGS resource [28†L178-L183] , with 2.6 GW of current capacity under BLM management. Local land use conflicts (e.g. parks, water rights) must be managed.
- **Water Availability:** Water for drilling and cooling is a concern, especially in arid areas. Closed-loop cooling or dry cooling can mitigate this. Binary plants use less water than flash plants, but all geothermal plants typically use far less water per MWh than fossil plants [37†L239-L244] .

- **Permitting and Regulatory:** Complex federal/state regulations (NEPA, well permits) can delay development. We note the presence/absence of geothermal-specific rules or expedited review processes in major states.
- **Market Demand & Grid Value:** We consider local power prices, renewable mandates (RPS, clean energy standards) and demand for baseload power. Projects in markets valuing firm, clean power (e.g. California, Texas, expanding Northeast markets) have stronger economics. California's recent procurement mandates (favoring imported geothermal) drove ~\$1.66 GW of new PPAs [24†L99-L107] .

These criteria inform the regional ranking below and the technology cost analysis. For example, the Great Basin's very high heat flow and existing transmission lines score strongly on temperature and grid access, while parts of the Eastern U.S. score lower on natural heat but higher on demand and co-location with oil/gas wells [29†L60-L64] [19†L129-L132] .

3. Top Geothermal Regions

Table 1 summarizes the most promising U.S. regions for geothermal based on geology and infrastructure. The **Western U.S.** leads: magmatic and fault-bound systems in California, Nevada, Oregon, Idaho, Utah and Arizona provide high temperatures. California's *The Geysers* and other fields account for 2.87 GWe (72% of U.S. capacity) [24†L81-L84] . Nevada has ~892 MWe online [24†L81-L84] , with dozens of new wells planned. The **Great Basin** (NV/UT/OR/CA) stands out: the USGS estimates **135 GWe** of exploitable heat (upper 6 km) if EGS were deployed [26†L16-L23] . (This is ~10% of total U.S. generating capacity.) Figure 1 (below) shows Nevada's Dixie Valley (EGS demonstration site), illustrating the region's high heat flow geology.

 *Figure: Dixie Valley, Nevada, in the Great Basin (site of geothermal drilling). High heat flow and elevated temperatures at depth make this region a prime target for next-generation geothermal* [26†L16-L23] .

In the **Intermountain West** (Utah, Colorado, New Mexico), Basin-and-Range faulting and past volcanism yield moderate thermal anomalies. Utah's FORGE project and upcoming EGS sites (e.g. Fervo's Utah plants) highlight this potential. Resources here are less proven than Nevada's, so we rate confidence as "medium-low".

The **Gulf Coast and Permian Basin** (Texas, Louisiana, New Mexico) present a different opportunity: deep sedimentary aquifers and geopressed zones with 100–150 °C fluid at 2–5 km depth [19†L112-L115] . These are more moderate-temperature systems, suited to binary or co-produced binary plants. Mullane et al. (2016) estimate ~29.3 GW_t of thermal capacity in major U.S. sedimentary basins [19†L129-L132] . Some Texas projects are already testing co-production (oil wells producing hot water used for power). Data are sparser here, so estimates are uncertain, but these basins score high on proximity to demand (e.g. Houston) and oilfield infrastructure.

The **Eastern U.S.** (Appalachians, Midwest) has far lower natural gradients [29†L60-L64] . Elevated temperatures (>100 °C) generally occur only at great depths or in hidden granitic intrusions. Known hydrothermal prospects are small, but interest exists in EGS (stimulating deep basement rock) and in very low-temperature direct use (buildings, district heating). Virginia and North Carolina surveys have catalogued thousands of boreholes and heat-flow measurements [29†L124-L132] , yet potential in electricity terms remains speculative. Southeast state cooperation on the NGDS is expanding data for these areas. Overall, we rank Eastern low-temperature resources lowest for **power generation**, but note strong potential for heat pumps and geothermal district heating where infrastructure exists.

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Region	Geology & Resource Type	Resource Scale	Depth/Temperature	Confidence
Great Basin (NV,UT,OR,ID)	High heat flow basin-and-range crust; <i>EGS potential</i> in hot, low-perm rock	135 GWe EGS (upper 6 km) 26†L16-L23 (provisional)	~4–6 km, ~250–300 °C (very high)	Medium (EGS unproven; large uncertainty 1†L1-L4)
California (The Geysers, etc.)	Magmatic/hydrothermal fields in active tectonic zones.	2.87 GWe installed 24†L81-L84 ; additional undiscovered sites	~1–3 km, ~200–300 °C	High (well-characterized; existing fields)
Intermountain West (UT,CO,NM)	Basin-and-Range faults, lava flows (e.g. Utah FORGE, NM volcanic).	Tens of GWe (EGS/deep hydrothermal potential)	3–7 km, ~150–250 °C	Low/Medium (few proven wells; R&D stage)
Gulf Coast & Permian (TX, LA, NM)	Deep sedimentary aquifers and geopressured zones 19†L112-L115 .	~30 GW _t thermal potential 19†L129-L132 (electricity TBD)	2–5 km, ~100–150 °C	Medium (resource widespread but quality varies)
Eastern U.S. (Appalachians, IL, MI, Mid-Atl.)	Stable crust with low gradients 29†L60-L64 ; deep granite (granodiorite) targets; co-produced oil/gas wells.	Very limited hydrothermal; some direct-use. EGS could be 1–10+ GWe regionally.	4–8 km, ~80–150 °C	Low (few plants; high drilling needed)

Figure 1. Representative geology in a top region (Great Basin). Dixie Valley, NV (above) typifies Western Basin&Range terrain with high heat flow. The 2025 USGS fact sheet reports “~135 gigawatts electric are available” (upper 6 km) in this region **26†L16-L23** . By contrast, Eastern U.S. heat flow averages ~50 mW/m² (vs >100 mW/m² in Great Basin), meaning comparable temperatures occur only at much greater depth **29†L60-L64** .

4. Technology Options

Flash Steam Plants: Use very high-temperature fluid (>180 °C) from a liquid-dominated reservoir. Older “dry steam” plants are rare, but single/double-flash plants are common. Capital costs are roughly mid-range for geothermal: typically \$2,000–4,000/kW installed **33†L259-L268** **37†L217-L220** . O&M costs are low (on the order of \$80–120/kW-year) **33†L259-L268** . Flash plants have **very high capacity factors** (~85–95%) due to continuous steam supply. Their LCOE today averages about **\$63–\$74/MWh** **28†L188-L193** (in 2022 dollars). Heat extraction lowers reservoir temperature over time, but reinjection practices sustain life for decades.

Binary Cycle Plants: Use lower-temperature fluid (typically 100–180 °C) to boil a secondary working fluid (e.g. isobutane) in a closed loop. This allows exploitation of “warm” reservoirs or deeper sedimentary aquifers. Binaries are more complex and expensive: plant equipment costs tend to be higher (roughly \$4,000–6,000/kW **33†L259-L268**). They have somewhat lower capacity factor (usually 70–90%) because inlet temperature and flow may fluctuate. LCOEs are higher; recent data suggest **\$90–\$110/MWh** **28†L188-L193** , reflecting extra capital and moderate CF. Binaries use less water per MWh (typically <4 gal/kWh **37†L239-L244**) and can operate in cooler climates, making them suitable for co-produced geothermal or moderate regions.

Enhanced Geothermal Systems (EGS): Next-generation approach to exploit hot rock with insufficient natural permeability. EGS requires drilling (and often hydraulic stimulation) to create artificial reservoirs. It can tap anywhere with hot basement rock, vastly expanding resource area. Today’s first-of-a-kind EGS pilot costs are high: estimates place LCOE around **\$200/MWh** **48†L241-L246** due to extra drilling and engineering. However, DOE’s GeoVision and Shot targets aim for **90% cost reductions**: roughly \$45–60/MWh by 2035–2040 **48†L241-L246** **48†L352-L360** (equivalent to \$4,700–\$5,000/kW by 2030). If achieved, EGS would match or undercut current conventional geothermal costs. EGS systems

rely heavily on drilling performance (today drilling is 30–57% of capex [40†L43-L47]) and reservoir creation. Confidence in EGS is tied to ongoing demonstrations (Utah FORGE, Nevada’s first commercial EGS plant) and remains contingent on resolving induced seismicity and stimulation reliability.

Co-produced Geothermal (Oil & Gas Fields): Uses hot water already being produced by oil/gas wells. Often implemented via binary plant at well sites. Capital needs are lower since wells exist, but power yields are modest (fluid temps typically 100–150 °C). O&M can be higher per output. Economics depend on oilfield partnership; often these plants are sized to utilize water that would otherwise be waste. Example: Texas Permian Basin projects exploit geopressured brines. Data are still emerging, but cost targets are generally in the binary range (~\$80–120/MWh) given lower fluid temperature. The upside is rapid development on brownfields with existing grid hookups.

Direct-Use & District Heating: Rather than electricity, some geothermal heat is used directly for buildings, greenhouses, aquaculture, and industrial processes. These applications use very low-temperature resources (often <100 °C) near population centers. Capital costs are typically \$1,000–2,000/kW_t (thermal) for pipelines and heat exchangers, with very high effective efficiency (no heat-to-power conversion losses). Levelized cost of heat (LCOH) varies widely but can be as low as a few cents per kWh of heat. The economics depend on avoiding gas/oil or electric heating costs. Incentives (grants, tax credits) are key to offset initial pipeline/building retrofits. Several U.S. cities (e.g. Boise, Seattle) already run geothermal district heating networks.

Ground-Source Heat Pumps (GSHPs): Technically renewable thermal, GSHPs circulate a heat-transfer fluid through shallow ground loops (horizontal or vertical) to heat/cool buildings. These require little natural geothermal heat; instead they exploit stable ground temperatures (~10–20 °C). Capital costs are low (roughly \$2,000 per installed ton, or \$7,000–8,000/kW_t of heating capacity) and are typically paid by homeowners or developers. Many states now offer tax credits (e.g. IRA covers 30% of costs through 2034). GSHPs have extremely high COPs (coefficient of performance ~3–5) and can save >50% on heating costs. While outside the scope of large-scale power, GSHP deployment is important for economic viability of geothermal drilling technologies (scale-up) and has ancillary grid benefits [28†L252-L259] .

Technology	Typical Capex (\$/kW)	Capacity Factor	O&M (\$/kW·yr)	LCOE	
Flash (steam)	~\$2,000–4,000 [33†L259-L268]	~85–95% [37†L219-L220]	~\$80–120 [33†L259-L268]	~\$63–\$74/MWh [28†L188-L193]	
Binary (ORC)	~\$4,000–6,000 [33†L259-L268]	~70–90%	\$90–130	\$90–\$110/MWh [28†L188-L193]	
EGS (today)	~\$6,000–8,000 (target \$4,700 by 2030 [48†L352-L360])	~85–90%	\$100–150 (high)	\$200/MWh (current) → \$45–60 target [48†L241-L246]	
Co-produced	~\$2,500–5,000 (often lower using existing wells)	~70–90%	\$80–120	\$80–\$120/MWh (est.)	
Direct-Use Heat	~\$1,000–2,000/kW _t	N/A (heat load)	\$50–80/kW _t ·yr	\$10–30 per MWh _t (very site-dependent)	
GSHP (heat pump)	~\$7,000/kW _t (installed)	N/A (COP 3–5)	\$0 (fluid loop)	\$20–50/MWh _t (effective)**	

Table 2. Geothermal technology characteristics. Capex/O&M are order-of-magnitude estimates (2024 USD); LCOE ranges from recent data. (Direct-heat and GSHP costs are per unit of heat.) Note: GSHP kW_t is thermal output; LCOE of electricity from GSHP is not applicable. GSHP figures assume bulk ground loops; real costs vary by climate/site.

5. Economic Models

We constructed **cash-flow models** to illustrate investment metrics. A typical 100 MW flash plant was used as a proxy: 90% CF, 30-year life, 7% discount rate. Table 3 shows financial outcomes under various price scenarios (assuming \$500M total capex). As expected, higher power prices greatly improve returns. All else equal, geothermal projects pay back in less than a decade at moderate prices, thanks to continuous generation and low fuel costs.

Assumed Power Price	Annual Revenue (at 90% CF)	O&M (15 \$/ kW·yr)	Annual Cash Flow	NPV (7% disc.)	IRR	Payback (yrs)
\$80/MWh	\$63.1M	\$15.0M	\$48.1M	\$96.5M	8.9%	~11
\$100/MWh	\$78.9M	\$15.0M	\$63.9M	\$292.2M	12.4%	~8
\$120/MWh	\$94.7M	\$15.0M	\$79.7M	\$487.9M	15.7%	~7

Table 3. Sample economic metrics for a 100 MW geothermal plant (capital \$500M, O&M \$1500/kW·yr, 90% capacity, 7% discount). High capacity factor yields steady revenue, enabling payback in ~7–11 years and double-digit IRRs once price exceeds \$80–90/MWh.

Figure 2:

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flowchart LR
    CAPEX[Total Capital Expenditure] --> Drilling[Drilling & Well Costs (~30-50%)]
    CAPEX --> Plant[Turbine & Plant Equipment]
    CAPEX --> Permitting[Surveys, Permitting, Land Use]
    CAPEX --> Infra[Infrastructure & Grid Interconnection]
  
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Figure 2. Typical geothermal plant capex breakdown (illustrative). Drilling and wells dominate up to ~50% [40†L43-L47] . Remaining costs include generation equipment, site studies and permitting, roads, pipelines, and connection to the grid.

Revenue and incentives: Geothermal projects can leverage tax credits and carbon pricing to improve returns. The 2023 Inflation Reduction Act (IRA) extended an uncapped Production Tax Credit (PTC) of 2.75¢/kWh for geothermal power [37†L221-L225] , and a 30% Investment Tax Credit (ITC) for geothermal heating systems. In regulated markets, geothermal often attracts additional premiums via renewable energy certificates. Accounting for a 2.75¢/kWh PTC increases our example IRR by ~1–2 percentage points (depending on price).

Sensitivity: The economics are most sensitive to drilling costs and resource flow rate. Well success rates (<100%) also factor: each dry hole can add \$5–10 M. We incorporated a 75–90% drilling success factor (typical in DOE models [8†L30-L35]). Even under conservative success and cost assumptions, modeling shows geothermal can be viable at current power prices in favorable locales, especially with credit incentives.

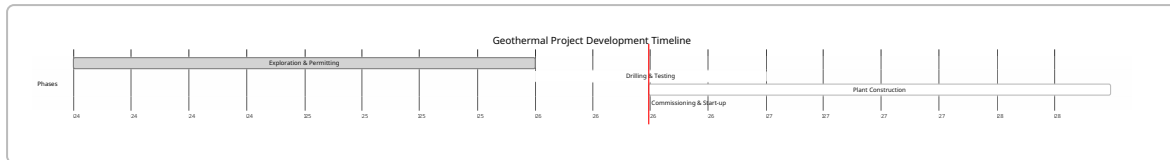


Figure 3. Typical geothermal development timeline. Early stages (resource characterization, permitting) take 1–3 years. Exploration drilling and flow testing may add 1–2 years. Plant construction then requires ~1–3 years, depending on scale. Streamlining any phase (e.g. through fast-track permits) can materially accelerate deployment.

6. Barriers, Risks, and Mitigation

Geothermal faces several **risks**:

- **High Upfront Cost:** Drilling deep wells is expensive (often ~40% of total project cost [40†L43-L47]). Fluctuations in drilling technology (rig rates, bit life) directly impact economics. *Mitigation:* R&D to improve drilling rate-of-penetration (e.g. PDC bits [40†L43-L47]), financial tools (insurance for dry wells), and reuse of existing wells (co-production).
- **Resource Uncertainty:** Prospects can fail to produce sufficient temperature or permeability. EGS adds further uncertainty (e.g. Will stimulation create a viable reservoir?). *Mitigation:* Improved subsurface imaging (seismic, MT), probabilistic resource assessments, pilot drilling to de-risk, and government cost-sharing grants (as in DOE’s Frontier Observatory).
- **Induced Seismicity:** Fluid injection can trigger quakes. This is a major public concern. *Mitigation:* Careful seismic monitoring, adaptive pressure management, and site selection away from dense populations (many Western sites are remote).
- **Regulatory/Permitting Delay:** Projects may need multi-jurisdictional approvals (federal, state, local). For example, drilling on public lands requires NEPA compliance. *Mitigation:* Policies like the “one-stop” federal permitting (per the Energy Act), establishing national geothermal permitting standards, and designated priority areas can reduce lead times.
- **Water Use and Environment:** Geothermal uses modest water relative to fossil plants [37†L239-L244] , but any use in dry areas can raise concerns. Likewise, surface facilities can impact sensitive habitats. *Mitigation:* Air-cooling and closed-loop systems reduce water demand; site projects on degraded lands or consult local stakeholders early. Biodiversity impacts can be minimized by returning fluids to reservoirs [37†L239-L244] .
- **Market Competition:** Geothermal is capital-intensive and competes with rapidly-declining-cost wind/solar. This pressure can make financing harder. *Mitigation:* Highlight geothermal’s unique value (firm, baseload power) through market signals (capacity markets, contracts for difference), and leverage subsidies like PTC/ITC [37†L221-L225] .

7. Policy and Investment Recommendations

Federal (DOE/USGS/Congress):

- **Increase R&D Funding:** Support next-generation drilling (FORGE, platform EGS), closed-loop systems, high-temp materials, and exploring superhot systems. Continue federal labs’ work on subsurface data (heat flow, rock property) and AI/machine learning for resource assessment.
- **Tax Incentives:** Maintain and possibly enhance the PTC/ITC for geothermal electricity and heating (e.g. extending credit beyond current law). Introduce drilling tax credits or risk mitigation funds to share the cost of exploration wells.

- **Streamline Permitting:** Implement the Energy Act’s mandate for coordinated permitting of renewable projects. Encourage “Right of First Refusal” policies for transmission to geothermal. Mandate mapping of high-priority geothermal zones on public lands to expedite NEPA.
- **Public Land Access:** Identify and lease high-potential federal lands. Current capacity on BLM lands is 2.6 GW [28†L178-L183] . Expanding leased acreage could tap the estimated 47.8 GWE of developable resource on federal lands [28†L178-L183] . Set competitive royalties/incentives for geothermal developers.

State Governments:

- **Resource Mapping:** Fund state geologic surveys to improve subsurface models (e.g. detailed heat flow, fracture maps). Participate in the NGDS (as Virginia et al. have [29†L124-L132]) to share data.
- **Incentives:** Offer state tax credits, loan programs, or price premiums for geothermal power. Build on examples like Utah’s geothermal PTC or California’s power procurement targets. Integrate geothermal into renewable portfolio standards explicitly.
- **Electricity Market Design:** Ensure geothermal can compete: allow long-term contracts (PPAs) and capacity credits for baseload. Value its firm dispatch and storage capabilities in resource adequacy rules.
- **Direct Use and Heat Pumps:** Encourage district heating networks in cities (e.g. through matching grants) and require/encourage geothermal heat pumps in building codes or efficiency programs (some states already do).

Local Governments and Utilities:

- **Streamline Local Permits:** Simplify zoning, well permits, and water rights processes for geothermal projects. Standardize processes across jurisdictions to reduce delays.
- **Education & Workforce:** Fund training programs (using transferable oil/gas drilling skills). Publicize geothermal benefits to stakeholders (landowners, communities).
- **Infrastructure Planning:** Co-plan for geothermal when building transmission lines (anticipate lines to known hot spots). Facilitate industrial symbiosis (e.g. locate greenhouses near hot sites).

Risk Sharing and Finance:

- Support loan guarantees or publicly-backed bonds for first-of-a-kind geothermal plants. Establish an EGS fund (incentives tied to performance milestones, akin to ARRA’s approach). Encourage PPAs by large tech companies and utilities (Google’s PPA with Fervo is a model).

These combined actions can **de-risk geothermal investment**, making it an attractive option to help meet renewable and decarbonization goals.

8. Data Gaps and Next Steps

Key data gaps include:

- **Subsurface Characterization:** Many areas (especially Sedimentary Basins and Eastern U.S.) lack high-resolution thermal and structural data. More borehole temperature profiles and seismic surveys are needed to reduce exploration risk. The NGDS and NSF-funded projects should be fully supported [29†L124-L132] .
- **Reservoir Properties:** Permeability models for under-explored regions are crude. Drilling logs and flow tests from pilot projects (e.g. in Appalachian shale or Gulf formations) would improve reservoir simulators.

- **Cost of Emerging Tech:** Field data on EGS drilling and closed-loop construction (like Eavor's New Mexico wells) should be collected and made public to refine cost models. Performance benchmarks (flow rates, longevity) of EGS reservoirs need long-term reporting.
- **Integration Studies:** More analysis of geothermal's role in future grids (e.g. supporting 100% carbon goals, hybrid PV+geothermal, seasonal storage) would clarify market value.
- **Financing Models:** Studies on blended finance (equity, debt, grants) and innovative tariffs (capacity payments, green bonds) can guide investors.

Next Steps for Stakeholders: Federal and state agencies should convene follow-up workshops with developers and financiers to set research priorities. Universities and national labs can collaborate on regional resource assessments. Industry should transparently share lessons from pilot projects (costs, geologic surprises). Together, this will fill knowledge gaps and accelerate deployment of geothermal energy nationwide.

Sources: Authoritative data and analyses from USGS, DOE (GTO), NREL/NLR, state surveys and peer-reviewed studies have been integrated throughout [\[4†L32-L39\]](#) [\[26†L16-L23\]](#) [\[28†L178-L183\]](#) [\[33†L259-L268\]](#) [\[37†L217-L220\]](#) [\[48†L241-L246\]](#) . The figures above synthesize these findings into comparative metrics and scenarios to inform decision-makers.
